TEACHER GUIDE

High school business

Revenue, costs and profit



A review lesson. Students run a car dealership in firms of four and review revenue, costs and profit on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min | **45 min** | 60 min | 75 min

What students do

Every student buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal inventory

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Total monthly costs

Round 1

A month at the lot.

Total costs for the month?

Fixed costs, a month	**$8,000**
You pay per car	**$17,500**
Cars sold	**4 cars**

ANSWER

$78,000

$8,000 + 4 × $17,500 = $78,000

Why: Fixed costs stay the same. Variable costs grow with every car.

2 Fixed or variable

Round 1

You list your dealership's costs for a month.

Which two are fixed costs?

- **A** Detailing each car sold
- **B** Rent on the showroom
- **C** Cars bought to sell
- **D** Commission on each sale
- **E** The manager's salary

ANSWER

✓ Rent on the showroom
✓ The manager's salary

Why: Rent and salaries stay the same however many cars you sell.

3 Rank by profit

Round 2

Three months: cars sold, the price, and total costs.

Order by profit, highest first?

- **A** 6 at $15,000, costs $75,000
- **B** 20 at $5,000, costs $90,000
- **C** 10 at $9,000, costs $70,000

ANSWER

1. 10 at $9,000, costs $70,000
2. 6 at $15,000, costs $75,000
3. 20 at $5,000, costs $90,000

Why: Profit: $20,000, $15,000, $10,000. The most revenue made the least profit.

Topic tasks

4 Average unit cost

Round 2

Cost of each car sold.

Average cost per car?

Fixed costs, a month	**$2,500**
Variable cost per car	**$9,500**
Cars sold	**5 cars**

ANSWER

$10,000

$2,500 ÷ 5 + $9,500 = $10,000

Why: Each car carries $500 of the fixed costs.

5 More sales or cuts

Round 3

Each car sold adds $400 profit. Sell 5 more a month, or cut fixed costs by $1,500 a month?

Which raises profit more, and why?

A Cut fixed costs by $1,500

B Sell 5 more cars

ANSWER

✔ **Sell 5 more cars**

Why: 5 × $400 = $2,000, more than the $1,500 saved.

6 Profit after rise

Round 3

Costs are going up.

Profit after the rise? Minus if below zero.

Sales	**$100,000**
Variable costs	**$45,000**
Fixed costs	**$15,000**
Variable costs rise by	**10%**

ANSWER

$35,500

$100,000 − $45,000 − 10% ($4,500) − $15,000 = $35,500

Why: The rise adds $4,500 to variable costs.

Final questions

Each student answers 6 final questions. They come from these 10 questions on cash, profit and sales tax.

1. You sell an EV city car to a customer for $11,000. In Indiana the customer pays 7% sales tax on top of the price. How much cash comes into your firm from this sale?
   ✔ **$11,770**

2. This round your firm sold cars to customers for $10,000, $17,000 and $21,000, and sold a Forklift to another dealer for $14,000. How much sales tax does the round's return ask for?
   ✔ **$3,360**

3. Your cash is $100,000 and your firm owes $2,100 of sales tax it has not filed yet. How does this affect the firm's net position?
   ✔ **It is $2,100 lower: unpaid sales tax is a debt and is subtracted**

4. Who pays the 7% sales tax on a car your firm sells to a customer?
   ✔ **The customer pays it on top of the price; your firm collects it and pays it to the Indiana Department of Revenue**

5. You bought a Panel van for $15,000 and sell it to a customer for $18,000. What is your gross profit on the van?
   ✔ **$3,000**

6. Your firm misses the round's sales tax return with $1,800 due. The late charge is 10% of the tax due, at least $5. What happens?
   ✔ **A $180 charge, and the $1,800 is still owed**

7. Cash $40,000, inventory at cost $30,000, investments $10,000, loans $25,000 and unpaid sales tax $1,400. What is the net position?
   ✔ **$53,600**

8. Why can a firm's cash look higher than the money it can really keep?
   ✔ **It holds sales tax collected from customers that belongs to the state until the return is filed**

9. A customer buys two Scooters 125cc at $3,900 each. How much sales tax does the customer pay?
   ✔ **$546**

10. Your firm collected $2,450 of sales tax this round and files the return on time. Cash before filing is $52,000. What is the cash after filing?
    ✔ **$49,550**